SÈRIE 1

Primera part

Exercici 1

Q1 a **Q2** c **Q3** c **Q4** d **Q5** b

Exercici 2

a)

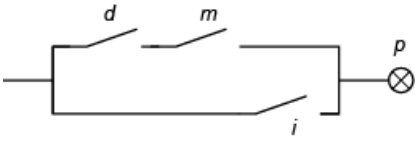
d	m	i	p
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

b) $p = (d + m + i)(d + \bar{m} + i)(\bar{d} + m + i)$

o també $p = (\bar{d} \cdot \bar{m} \cdot i) + (\bar{d} \cdot m \cdot i) + (d \cdot \bar{m} \cdot i) + (d \cdot m \cdot \bar{i}) + (d \cdot m \cdot i)$

$p = d \cdot m + i$

c)



Segona part

OPCIÓ A

Exercici 3

a) $P_{\text{vent}} = \frac{1}{2} A \cdot \rho \cdot v^3 = 955{,}2 \text{ kW}$

b) $P_{\text{útil}} = P_{\text{vent}} \cdot c_{\text{a}} \cdot \eta_{\text{aerog}} = 272{,}8 \text{ kW}$

c) $E_{\text{total}} = P_{\text{útil}} \cdot t \cdot n = 73{,}66 \cdot 10^9 \text{Wh} = 265{,}2 \cdot 10^{12} \text{ J}$

d) $\text{Ingressos} = p_{\text{venda}} \cdot E_{\text{total}} = 5616 \text{ k€}$

e) $\text{Reducció} = (1 - 0.9^3) \cdot 100 = 27{,}1\%$

Exercici 4

a) $\Gamma_{\text{motor}} = \dfrac{P_{\text{subm}}}{\omega} = 2{,}876 \text{ Nm}$

b) $v_{\text{pat}} = \omega \cdot d_{\text{roda}}/2 = 5{,}563 \text{ m/s} = 20{,}03 \text{ km/h}$

c) $E_{\text{útil}} = E_{\text{bat}} \cdot \eta_{\text{mot}} = 222{,}5 \text{ Wh} = 801{,}0 \text{ kJ}$

d) $t_{\text{màx}} = \dfrac{E_{\text{útil}}}{P_{\text{subm}}} = 4005 \text{ s} = 1{,}113 \text{ h}$; $\quad s_{\text{màx}} = v_{\text{pat}} \cdot t_{\text{màx}} = 22{,}28 \text{ km}$

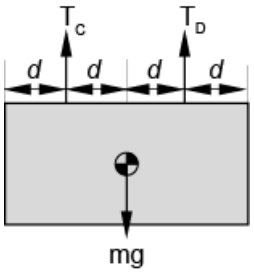
OPCIÓ B

Exercici 3

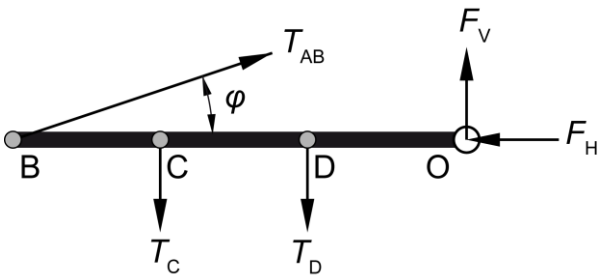
a) $\sum F_{\text{verticals}} = 0 \;\rightarrow\; T_C + T_D = mg$

$\sum M(G) = 0 \;\rightarrow\; T_C \cdot d - T_D \cdot d = 0 \;\rightarrow$

$T_C = T_D = \dfrac{mg}{2} = 58{,}84 \text{ N}$



b)



c) $\varphi = \text{atan}\left(\dfrac{1}{3}\right) = 18{,}43°$

d) $\sum M(O) = 0$; $\quad -T_{AB} \sin\varphi \cdot 3L + T_C \cdot 2L + T_D \cdot L = 0 \quad \rightarrow \quad T_{AB} = 186{,}1 \text{ N}$

e) $\sum F_{\text{horitzontals}} = 0 \;\rightarrow\; F_H = T_{AB} \cdot \cos\varphi \quad \rightarrow \quad F_H = 176{,}5 \text{ N}$

$\sum F_{\text{verticals}} = 0 \quad \rightarrow \quad T_{AB} \sin\varphi + F_V = T_C + T_D \quad \rightarrow \quad F_V = 58{,}84 \text{ N}$

Exercici 4

a) $P_{\text{útil}} = \dfrac{E_{\text{útil}}}{t} = \dfrac{m \cdot c_{\text{e}} \cdot \Delta T}{t} = \dfrac{\rho_{\text{aigua}} \cdot V \cdot c_{\text{e}} \cdot \Delta T}{t} = \rho_{\text{aigua}} \cdot q_{\text{aigua}} \cdot c_{\text{e}} \cdot \Delta T$

$$q_{\text{aigua}} = \frac{P_{\text{útil}}}{\rho_{\text{aigua}} \cdot c_{\text{e}} \cdot \Delta T} = 16{,}08 \text{ L/min}$$

b) $P_{\text{cons}} = \dfrac{P_{\text{útil}}}{\eta_{\text{c}}} = 32{,}18 \text{ kW}; \quad q_{\text{comb}} = \dfrac{P_{\text{cons}}}{p_{\text{c}}} = 0{,}5191 \text{ g/s}$

c) $t = \dfrac{V}{q_{\text{aigua}}} = 6{,}22 \text{ min}; \quad m = t \cdot q_{\text{comb}} = 193{,}7 \text{ g}$